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## SAMPLE QUESTION PAPER

Course - DAC Module - C++ - DS



Q. No. 1

In following code what will be the state of c1 object.

```
class Complex
```

```
{
```

```
int real, imag;
```

```
public:
```

```
Complex()
```

```
{
```

```
Complex( 10, 20 );
```

```
}
```

```
Complex( int real, int imag )
```

```
{
```

```
this->real = real;
```

```
this->imag = imag;
```

```
}
```

```
friend ostream& operator<<( ostream& out, const Complex& other )
```

```
{
```

```
out<<"Real : "<<other.real<<endl;
```

```
out<<"Imag : "<<other.imag<<endl;
```

```
return out;
```

```
}
```

```
};
```

```
int main( void )
```

```
{
```

```
static Complex c1;
```

```
cout<<c1<<endl;
```

```
return 0;
```

```
}
```

A. 10, 20      B. Compile Time Error      C. 0,0      D. Garbage Value

Q. No. 2

In C/C++ language '3' represents -

A. A digit      B. An Integer      C. A Character      D. A Word

Q. No. 3

Default mode of inheritance in C++ is -

A. Private

B. Public

C. Protected

D. C++ do not support mode of inheritance

5) whole darn of run in place of call. So it can be Runtime or compile time



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Q No. 4

Object Oriented Programming -

- A. Makes Program More reliable
- B. Simulates real life.
- C. Uses a lot of intimidating vocabulary, which is not as bad as it sounds.
- D. All of the above.

It is used to simplify complex logic of real time problems using top down approach.

6)

Person

|      |
|------|
| Name |
| Age  |

Q No. 5

Inline functions are invoked at the time of -

- A. Run time
- B. Compile Time
- C. Depends on how it is invoked
- D. Both B and C above

Inline are used to reduce construction of function activation records. They just place

Q No. 6

What is an aggregate object? What is an aggregate object?

- A. An object instance that has only static methods.
- B. An object instance that has only primitive attributes.
- C. An object instance that contains other objects.
- D. An object that has only primitive attributes and instances methods.

Address

|      |
|------|
| id   |
| Slot |

Q No. 7

Which of the following is not a valid conditional inclusions in preprocessor directives

- A. #ifdef
- B. #ifndef
- C. #endif
- D. #elif

Q No. 8

Which of the following operators can be implemented as a nonmember operator?

- A. =
- B. ( )
- C. [ ]
- D. <<

because of stream operators are private.

Q No. 9

If we treat an object as a function then it is called -

- A. Function Pointer
- B. Smart Pointer
- C. Function Object
- D. Iterator

long("sunbeam")

Q No. 10

A friend of a class is also automatically a friend of

- A. Its base class.
- B. Its derived class.
- C. Any other friends of the class
- D. All of the above
- E. None of the above

transitivity rule

Q No 11

In which mode of inheritance we can access private members inside derived class?

- A. private
- B. protected
- C. public
- D. None of the above

10) If B is friend of A and C is friend of B then C is not friend of A



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Q No 12

In following code what should be state of derivedInstance?

```
#include<iostream>
using namespace std;
class Base
{
private: int num1;
Base( )
{
this->num1 = 10;
}
Base( int num1 )
{
this->num1 = num1;
}
};
class Derived : public Base
{
int num2;
public: Derived( int num1, int num2 ) : Base( num1 )
{
this->num2 = num2;
}
};
int main( void )
{
Derived derivedInstance( 500, 600 );
return 0;
}
```

A. 500 600

B. Compiler error.

C. 10 600

D. None of the above.

Q No. 13

What is the output of following program

```
#include<iostream>
using namespace std;
class A
{
public:
int num1;
public: A()
{
this->num1 = 10;
}
};
class B : public A
{
public:
```



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```
int num2;
public: B()
{
    this->num1 = 20;
    this->num2 = 30;
};
class C : public A
{
    public: int num3;
    public: C()
    {
        this->num1 = 40; this->num3 = 50;
    };
};
class D : public B, public C
{
    public: void Print()
    {
        cout<<B::num1<<" "<<num2<<" "<<num3<<endl;
    };
};
int main( void )
{
    D obj;
    obj.Print();
    return 0;
}
```

A → B → A → C → D

- A. 20 30 50  
B. 40 30 50

- C. 30 40 50  
D. Ambiguity Error

Q No. 14 Which of the following member inherit into derived class

- A. Friend class  
B. static member function  
C. Assignment operator function  
D. both A and C

Q No 15 - Process of converting reference / pointer of parent type into child type is called

- A. Widening  
B. Upcasting  
C. Narrowing  
D. Downcasting  
E. Shadowing





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Q No. 16 -

What is the output of following code

```
class Base
{
public:
void Print( void )
{
cout<<"Base::Print"<<endl;
}
};
class Derived : Base
{
public:
virtual void Print( void )
{
cout<<"Derived::Print"<<endl;
}
};
int main( void )
{
Base* ptr = new Derived();
ptr->Print();
return 0;
}
```

A. Base::Print

B. Derived::Print

C. Compiler error

D. None of the above.

Q No. 17 If all members of the class are pure virtual then it is called as

A. Concrete class

B. Abstract class

C. Pure virtual class

D. Interface

Q No 18 -With the help of runtime polymorphism we can achieve

a. Reusability.

b. Modularity.

*Product of polymorphism*

c. Maintainability.

d. Reliability.

Q No 19 -What does the following declaration mean: int ( \*ptr )[ 10 ];

A. ptr is array of pointers to 10 integers

B. ptr is pointer to an array of 10 integers

C. ptr is an array of 10 integers

D. ptr is pointer to array

Q No 20 - wchar\_t datatype is -

A. Macro having value unsigned int

B. typedef of unsigned int

C. Macro having value unsigned short

D. typedef of unsigned short



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Q No 21

What is the output of following code

```
#include<iostream>
using namespace std;
int main( void )
{
    char* ptr = "SunBeam";
    cout<<*ptr++<<" "<<*ptr<<endl;
    return 0;
}
```

- A. S u
- B. u n
- C. S n
- D. u u

Q No 22 Find out the output

```
class Base
{
public:
    virtual void Print( void )
    {
        cout<<"Base::Print"<<endl;
    }
};
class Derived : public Base
{
private:
    virtual void Print( void )
    {
        cout<<"Derived::Print"<<endl;
    }
};
int main( void )
{
    Base* ptr = ( Base* )new Derived();
    ptr->Print();
    return 0;
}
```

- A. Base::Print
- B. Compiler Error : function is private in derived class
- C. Derived::Print
- D. Compiler Error : Can not cast Derived\* to Base\*

Q 23 To find out true type of object, If we use null pointer with typeid then output is

- A. bad\_cast exception
- B. bad\_alloc exception
- C. bad typeid exception
- D. bad\_pointer exception

*to find out true type of object using null pointer for C++ runtime throws bad typeid exception*



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Q No. 24 In case of polymorphic type, if we want to do downcasting then we should use

- A. static\_cast operator
- B. dynamic cast operator
- C. const\_cast operator
- D. reinterpret\_cast operator

Q No 25 Which stream class is used to both read and write on files ?

- A. fstream
- B. ifstream
- C. ofstream
- D. iostream

Q No 26 Consider Stack implemented using array

```
#define MAX 10
```

```
struct Stack
```

```
{  
    int top; //top = -1  
    int arr[ MAX ]; };
```

In this implementation of stack maximum value of top which can not cause overflow will

- A. 11
- B. Any other
- C. 10
- D. 9

Q No 27 Suppose we need to sort a list of employee records in ascending order, using the social security number (a 9-digit number) as the key (i.e., sort the records by social security number). If we need to guarantee that the running time will be no worse than  $n \log n$ , which sorting methods could we use?

- A. Merge sort
- B. Quick sort
- C. Insertion sort
- D. Either Mergesort or Quicksort
- E. None of these

Q No 28 Consider a class List that implements an unordered list. Suppose it has as its representation a singly linked list with a head and tail pointer. Given that representation, which of the following operations could be implemented in  $O(1)$  time?

- I Insert item at the front of the list
- II Insert item at the rear of the list
- III Delete front item from list
- IV Delete rear item from list

- A. I and II
- B. I and III
- C. I, II, and III
- D. I, II, and IV
- E. all of them

Q No 29 Binary search algorithm cannot be applied to

- A. sorted linked list
- B. sorted binary trees
- C. sorted linear array
- D. pointer array

Q No. 30 A vertex of graph is called even or odd depending upon

- A. Total number of edges in graph is even or odd
- B. Total number of vertices in graph is even or odd



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- C. Its degree is even or odd  
D. None of these.

Q No. 31 The depth of a complete binary tree is given by

- A.  $D_n = n \log_2 n$   
B.  $D_n = n \log_2 n + 1$   
C.  $D_n = \log_2 n$   
D.  $D_n = \log_2 n + 1$

Q No. 32 The degree of any vertex of graph is .... ?

- A. The number of edges incident with vertex  
B. Number of vertex in a graph  
C. Number of vertices adjacent to that vertex  
D. Number of edges in a graph

Q No. 33 - What is the best definition of a collision in a hash table?

- A. Two entries are identical except for their keys.  
B. Two entries with different data have the exact same key.  
C. Two entries with different keys have the same exact hash value.  
D. Two entries with the exact same key have different hash values.

Q No. 34 In a graph, when does Dijkstra's algorithm stop?

- A. When the shortest path to the destination vertex is found  
B. When all the vertices in the graph are included to the path  
C. When the vertices together form a cycle  
D. When the minimum spanning tree is constructed

Q No. 35 Backtracking algorithms determine problem's solutions by  
.....searching the solution space for the given problem instance

- A. Logically  
B. Periodically  
C. Non- Systematically  
D. Systematically

Q No. 36 The time taken by nondeterministic sorting algorithm is \_\_\_\_\_

- A.  $O(1)$   
B.  $O(\log n)$   
C.  $O(n^2)$   
D.  $O(n \log n)$   
E.  $O(n)$

Q No. 37 What does the size of the vector refers to in c++?

- A. Size of vector  
B. Type of vector  
C. Number of elements  
D. None of the mentioned



size = 3

Capacity



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Module – C++ - DS



Q No. 38 - What is shallow copy?

- A. A shallow copy creates a copy of the dynamically allocated objects too.
- B. A shallow copy just copies the values of the data as they are.
- C. A shallow copy creates a copy of the statically allocated objects too
- D. Both b and c above

Q No. 39 \_\_\_\_\_ is the process of compartmentalizing the elements of an abstraction that constitute its structure and behavior?

- A. Abstraction
- B. Encapsulation
- C. Modularity
- D. Hierarchy

Q No. 40 \_\_\_\_\_ is the property that distinguishes an active object from one that is not active?

- A. Typing
- B. Concurrency
- C. Persistence
- D. Threading

{

c  
c

c  
t

Q  
t

Q

--



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## SAMPLE QUESTION PAPER

Course – DAC

Module – Operating System Concepts



- Q No 1 Which bash command is used to do floating number mathematical operations?
- A. expr C. bc  
B. calc D. echo
- Q No 2 BASH shell stands for?
- A. Bourne-again Shell C. Basic to Advanced Shell  
B. Basic Access Shell D. Big & Advanced Shell
- Q No 3 Which of the following file will not be displayed by the command "ls re\*"  
" ?
- A. reminder C. Receipt ✓  
B. receipt D. recipe-cake
- Q No 4 Shell is ?
- A. Command Interpreter C. Interface between user and applications  
B. Interface between Kernel and Hardware D. Command Compiler
- Q No 5 Process is \_\_\_\_\_
- A. A program in execution.  
B. An instance of a program running on a computer.  
C. The entity that can be assigned to and executed  
D. All of the above.
- Q No 6 \_\_\_\_\_ is a large kernel, including scheduling file system, networking, device drivers, memory management and more.
- A. Monolithic kernel C. Macro kernel  
B. Micro kernel D. Mini kernel
- Q No 7 \_\_\_\_\_ Involves treating main memory as a resource to be allocated to and shared among a number of active processes.
- A. Partition management C. Disk management  
B. Memory management D. All of the above
- Q No 8 A process that execute only in main memory is referred to as \_\_\_\_\_ and that allocated in disk is referred to a \_\_\_\_\_
- A. virtual memory, true memory C. real memory, virtual memory  
B. virtual memory, real memory D. imaginary memory, real memory
- Q No 9 The sum of the seek time, and the rotational delay is called the \_\_\_\_\_
- A. reached time  
B. access time  
C. arrived time  
D. common time



## SAMPLE QUESTION PAPER

Course – DAC      Module – Operating System Concepts



- Q No 10 In process scheduling \_\_\_\_\_ determines which ready process will be executed next by processor.
- A. long term scheduling
  - B. medium term scheduling
  - C. short term scheduling
  - D. none of the above
- Q No 11 The \_\_\_\_\_ determines when a page should be brought into main memory.
- A. Fetch policy
  - B. Placement policy
  - C. Replacement policy
  - D. Resident set management
- Q No 12 Which command will set only read permission to others , only write permission to user and only execute permission to group for any file ?
- A. filename 777 chmod
  - B. chmod 437 filename
  - C. chmod 214 filename
  - D. chmod 421 filename
- Q No 13 Process Management function of an operating system kernel includes.
- A. Process creation and termination.
  - B. Process scheduling and dispatching
  - C. Process switching
  - D. All of the above
- Q No 14 \_\_\_\_\_ is the ability of multiple process to co-ordinate their activities by exchange of information
- A. Synchronization
  - B. Mutual Exclusion
  - C. Dead lock
  - D. Starvation
- Q No 15 \_\_\_\_\_ is a high level abstraction over Semaphore.
- A. Shared memory
  - B. Message passing
  - C. Monitor
  - D. Mutual exclusion
- Q No 16 The producer-consumer problem refers to:
- A. Mutual-exclusion when n jobs sharing a resource
  - B. Multiple clients and servers accomplishing a computation
  - C. The need for synchronization in systems where many processes share a resource
  - D. None of the above





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Course – DAC      Module – Operating System Concepts



Q No 17 Thread-specific data is data that...

- A. is not associated with any process.
- B. has been modified by the thread but not yet updated to the parent process.
- C. is generated by the thread independent of the thread's process .
- D. is copied and not shared with the parent process.

Q No 18 User programs and system services interact in a microkernel architecture by using \_\_\_\_\_

- A. signal
- B. Interprocess Communication Mechanisms
- C. interrupts
- D. None of the above

Q No 19 Inter process communication:

- A. Allows processes to communicate and synchronize their actions when using the same address space.
- B. Allows processes to communicate and synchronize their actions without using the same address space.
- C. Allows the processes to only synchronize their actions without communication.
- D. None of these

Q No 20 The removal of process from active contention of CPU and reintroduce them into memory later is known as \_\_\_\_\_

- A. Interrupt
- B. Swapping
- C. Signal
- D. Multiprogramming

Q No 21 What is the memory from 1K - 640K called?

- A. Extended Memory
- B. Normal Memory
- C. Low Memory
- D. Conventional Memory

Q No 22 Which is not the state of the process?

- A. Blocked
- B. Running
- C. Ready
- D. Privileged

Q No 23 What is true about touch command?

- A. it changes the time stamp of existing file
- B. it creates multiple empty files at once
- C. option A is true
- D. All of above

Q No 24 The state of a process after it encounters an I/O instruction is \_\_\_\_\_

- A. Ready
- B. Blocked/Waiting
- C. Idle
- D. Running



# SAMPLE QUESTION PAPER

Course – DAC Module – Operating System Concepts



- Q No 25 Which directory implementation is used in most Operating System?
- A. Single level directory structure
  - B. Two level directory structure
  - C. Tree directory structure
  - D. Acyclic directory structure
- Q No 26 Which of the following is NOT a way to make file systems faster?
- A. Put parts of a file on many different tracks so part of it can be accessed no matter where the disk head is
  - B. Cache frequently used blocks in memory so they can be accessed at memory speeds instead of disk speeds
  - C. Use a disk scheduling algorithm to minimize the distance between seeks
  - D. Put frequently used files or directories near the center of the disk so on average they won't be far from the read head
- Q No 27 Which one of the following is a synchronization tool?
- A. thread
  - B. pipe
  - C. semaphore
  - D. socket
- Q No 28 When several processes access the same data concurrently and the outcome of the execution depends on the particular order in which the access takes place, is called
- A. dynamic condition
  - B. race condition
  - C. essential condition
  - D. critical condition
- Q No 29 Which one of the following explains the sequential file access method?
- A. random access according to the given byte number
  - B. read bytes one at a time, in order
  - C. read/write sequentially by record
  - D. read/write randomly by record
- Q No 30 The usefulness of signals as a general inter process communication mechanism is limited because :
- A. they do not work between processes
  - B. they are user generated
  - C. they cannot carry information directly
  - D. None of these
- Q No 31 For each file there exists a \_\_\_\_\_, that contains information about the file, including ownership, permissions and location of the file contents.
- A. metadata
  - B. file control block
  - C. process control block
  - D. All of these
- Q No 32 In UNIX, the return value for the fork system call is \_\_\_\_\_ for the child process and \_\_\_\_\_ for the parent process.
- A. A Negative integer, Zero
  - B. Zero, A Negative integer
  - C. Zero, A nonzero integer
  - D. A nonzero integer, Zero



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Course – DAC Module – Operating System Concepts



Q No 33 The following C program : `main() { fork();fork();printf( "yes" ); }` prints yes:

- A. only once
- B. twice
- C. four times
- D. eight times

Q No 34 How will you copy the current directory file names in a new file?

- A. ls > dircontent.txt
- B. cat > dircontent.txt
- C. touch dircontent.txt
- D. touch ls -l

Q No 35 Which of the following call never returns an error ?

- A. getpid ✓
- B. fork
- C. ioctl — to perform certain control line
- D. open

Q No 36 How do you see hidden files in your current directory?

- A. ls -h
- B. ls -l -a
- C. ls -s -l
- D. ls -R -t

Q No 37 Which command is used to display all the files including hidden files in your current and its subdirectories ?

- A. ls -a -R
- B. ls -a
- C. ls -R
- D. ls -l

Q No 38 What would be the current working directory at the end of the following command sequence?

```
> pwd
/home/user1/proj
> cd src
> cd generic
> cd
> pwd
```

- A. /home/user1/proj
- B. /home/user1/proj/src
- C. /home/user1
- D. /home/user1/proj/src/generic

Q No 39 The statement `z='expr 5 / 2'` would store which of the following value in z?

- A. 0
- B. 1
- C. 2
- D. 2.5
- E. 3

Q No 40 Create a new file new.txt that is a concatenation of contents of file1.txt and file2.txt

- A. cp file.txt file2.txt new.txt
- B. cat file1.txt file2.txt > new.txt
- C. mv file[12].txt new.txt
- D. ls file1.txt file2.txt | new.txt

Q  
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---



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## SAMPLE QUESTION PAPER

Course – DAC      Module – DBT



Q No 1 SQL> desc countries

| NAME         | NULL?    | TYPE         |
|--------------|----------|--------------|
| COUNTRY_ID   | NOT NULL | CHAR(2)      |
| COUNTRY_NAME |          | VARCHAR2(40) |
| REGION_ID    |          | NUMBER       |

Which of the following SQL statements will correctly change the value of REGION\_ID to 4, for all records, where it is currently set to 2?

- A. UPDATE countries region\_id='4' WHERE region\_id HAVING 2;
- B. UPDATE countries SET region\_id=4 WHERE region\_id='2';
- C. UPDATE countries SET region\_id='4';
- D. UPDATE countries region\_id='4' WHERE region\_id=2;

Q No 2 Which of the following feature is not true for INTERSECT operator?

- A. On reversing the order of the intersected table it does not alter the result.
- B. The INTERSECT operator ignores the NULL values that are present in both the tables that are being intersected.
- C. The number of columns and data types must be identical for all SELECT statements in the query containing the INTERSECT operator.
- D. The names of columns in all SELECT statements containing the INTERSECT operator may or may not be identical.

Q No 3 View the following table descriptions:

SQL> desc sales

| Name          | Null?    | Type         |
|---------------|----------|--------------|
| PROD_ID       | NOT NULL | NUMBER       |
| CUST_ID       | NOT NULL | NUMBER       |
| TIME_ID       | NOT NULL | DATE         |
| CHANNEL_ID    | NOT NULL | NUMBER       |
| PROMO_ID      | NOT NULL | NUMBER       |
| QUANTITY_SOLD | NOT NULL | NUMBER(10,2) |
| AMOUNT_SOLD   | NOT NULL | NUMBER(10,2) |

SQL> desc customers

| Name                   | Null?    | Type         |
|------------------------|----------|--------------|
| CUST_ID                | NOT NULL | NUMBER       |
| CUST_FIRST_NAME        | NOT NULL | VARCHAR2(20) |
| CUST_LAST_NAME         | NOT NULL | VARCHAR2(40) |
| CUST_GENDER            | NOT NULL | CHAR(1)      |
| CUST_STREET_ADDRESS    | NOT NULL | VARCHAR2(40) |
| CUST_POSTAL_CODE       | NOT NULL | VARCHAR2(10) |
| CUST_CITY              | NOT NULL | VARCHAR2(30) |
| CUST_STATE_PROVINCE    | NOT NULL | VARCHAR2(40) |
| COUNTRY                | NOT NULL | VARCHAR2(40) |
| CUST_MAIN_PHONE_NUMBER | NOT NULL | VARCHAR2(25) |
| CUST_EMAIL             |          | VARCHAR2(30) |



# SAMPLE QUESTION PAPER

Course – DAC      Module – DBT



.....  
Which SQL statement provides a list of total sales by CUSTOMER\_ID for customers with total individual sales orders greater than 280?

A. SELECT b.cust\_id, COUNT(\*)  
FROM sales a, customers b  
WHERE a.cust\_id=b.cust\_id  
GROUP BY b.cust\_id  
HAVING COUNT(\*) > 280;

C. SELECT b.cust\_id, COUNT(\*)  
FROM sales a, customers b  
WHERE a.cust\_id=b.cust\_id  
GROUP BY b.cust\_id;

B. SELECT b.cust\_id, COUNT(\*)  
FROM sales a, customers b  
WHERE a.cust\_id=b.cust\_id  
GROUP BY b.cust\_id  
HAVING AVG(\*) > 280;

D. SELECT b.cust\_id, AVG(\*)  
FROM sales a, customers b  
WHERE a.cust\_id=b.cust\_id  
GROUP BY b.cust\_id  
HAVING AVG(\*) > 280;

Q No 4 Which of the following is the invalid uses of the transaction control statements?

- A. End a transaction by making the changes permanent or undoing the changes
- B. Explicitly begin a transaction, choosing statement-level consistency or transaction-level consistency
- C. Determine whether an update or insert should be executed
- D. Set undo savepoints and undo changes back to a savepoint

Q No 5 Which of the following is the characteristics of materialized views?

- A. They require a dedicated network connection for a materialized view.
- B. They consume storage space and also used to replicate data from another database.
- C. They used in data-warehouse environments where data needs to be aggregated and stored so that queries and reports run faster.
- D. Both B and C

Q No 6 Which of the following can be used to populate a table?

- A. MERGE statement
- B. INSERT statement
- C. SQL\*Loader
- D. All of the above

Q No 7 A Cartesian product generates a large number of rows, and the result of a Cartesian product is rarely used. This can be considered a disadvantage of a Cartesian product.

Which of the following are the reasons for using a Cartesian product?

- A. It is used to retrieve the desired rows from a table.
- B. It is used to join two tables & to simulate a reasonable amount of data.
- C. It is used to retrieve all rows from a table.
- D. None of the above



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Q No 8 The database contains a table named Job\_recruitment. The table contains columns such as Joining\_date, Leaving\_date, Tot\_salary, and Tot\_incentive. The Joining\_date and Leaving\_date columns are of DATE data type. You perform the following arithmetic operation on these columns:

Joining\_date - Leaving\_date

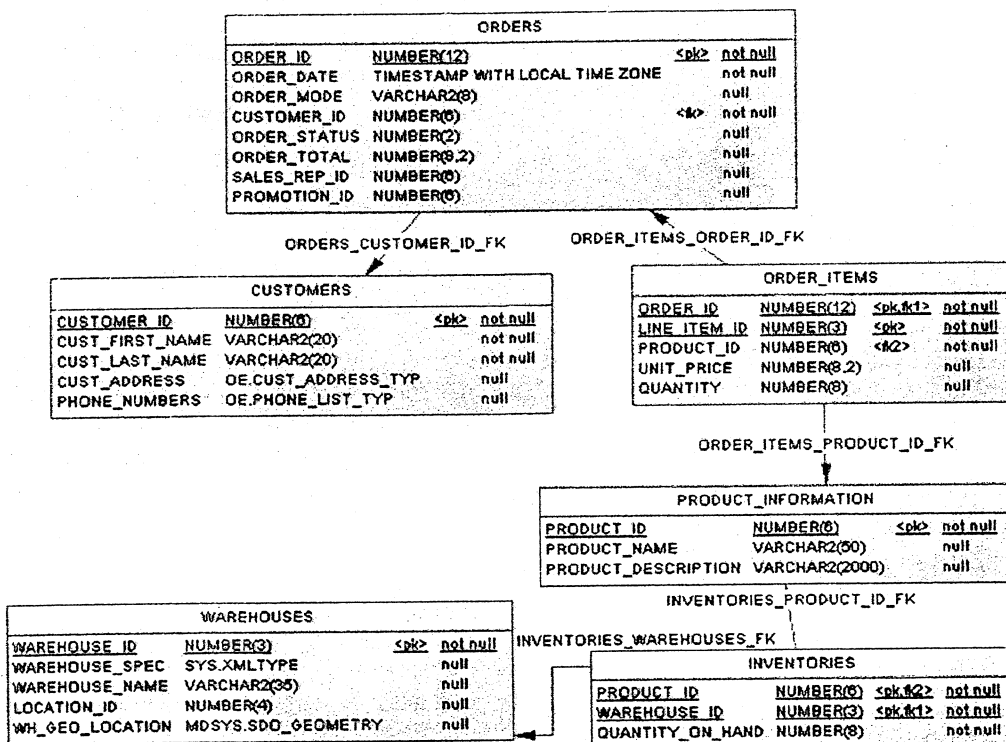
What will the above arithmetic operation return?

- A. It will return an ORA error.
- B. It will return the result in NUMBER data type.
- C. It will return the result in VARCHAR2 data type.
- D. It will return the result in DATE data type.

Q No 9 You work as a Database Designer for XYZ Inc. The company uses Oracle 11g as its database. The database contains a table named Employees. You want to retrieve the first name and address of those employees who earn a salary of INR10000. The Emp\_salary column is of NUMBER(8,2) data type. Which of the following SQL statements will you use to accomplish the task?

- A. SELECT Emp\_first\_name, Emp\_add FROM Employees;
- B. SELECT Emp\_first\_name, Emp\_add FROM Employees WHERE Emp\_salary = "10000";
- C. SELECT Emp\_first\_name, Emp\_add FROM Employees WHERE Emp\_salary = "INR10000";
- D. SELECT Emp first name, Emp add FROM Employees WHERE Emp salary = 10000;

Q No 10 Review the following ER diagram:



Which of the given SQL statements will do the following tasks?



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1. Display the following columns:  
    cust\_first\_name and cust\_last\_name from CUSTOMERS table.  
    order\_date, order\_mode, order\_status, and order\_total from ORDERS table
2. Order by the customer\_id column in CUSTOMERS
3. Display both customers with and without orders in the ORDERS table
- A. SELECT a.cust\_first\_name, a.cust\_last\_name, b.order\_date, b.order\_mode, b.order\_status, b.order\_total  
    FROM customers a, orders WHERE a.customer\_id=b.customer\_id ORDER BY a.customer\_id;
- B. SELECT a.cust\_first\_name, a.cust\_last\_name, b.order\_date, b.order\_mode, b.order\_status, b.order\_total  
    FROM customers a, orders b WHERE a.customer\_id(+) = b.customer\_id ORDER BY a.customer\_id;
- C. SELECT a.cust\_first\_name, a.cust\_last\_name, b.order\_date, b.order\_mode, b.order\_status, b.order\_total FROM customers a, orders b WHERE customer\_id=customer\_id (+) ORDER BY a.customer\_id;
- D. SELECT a.cust first name, a.cust last name, b.order date, b.order mode, b.order status, b.order total FROM customers a, orders b WHERE a.customer id=b.customer id (+) ORDER BY a.customer id;
- E. SELECT a.cust\_first\_name, a.cust\_last\_name, b.order\_date, b.order\_mode, b.order\_status, b.order\_total  
    FROM customers a, orders b ORDER BY a.customer\_id;

Q No 11 The PQR Inc. company uses an Oracle 11g database. The database contains two tables named Employees and Employee\_ID. You are required to write a query that will include a subquery. The subquery will be returning multiple rows. Which of the following operators can you use with the subquery?

Each correct answer represents a complete solution. Choose all that apply.

- i. LIKE
  - ii. NOT IN
  - iii. BETWEEN
  - iv. IN
- A. Only i
  - B. Only ii
  - C. Only i and iii
  - D. Only ii and iv

Q No 12 Which of the following is NOT a database object?

- A. Synonym
- B. Sequence
- C. View
- D. Data file

Q No 13 You have issued the CREATE TABLE command to create a table named Employee. You use the following code-line to accomplish the task:





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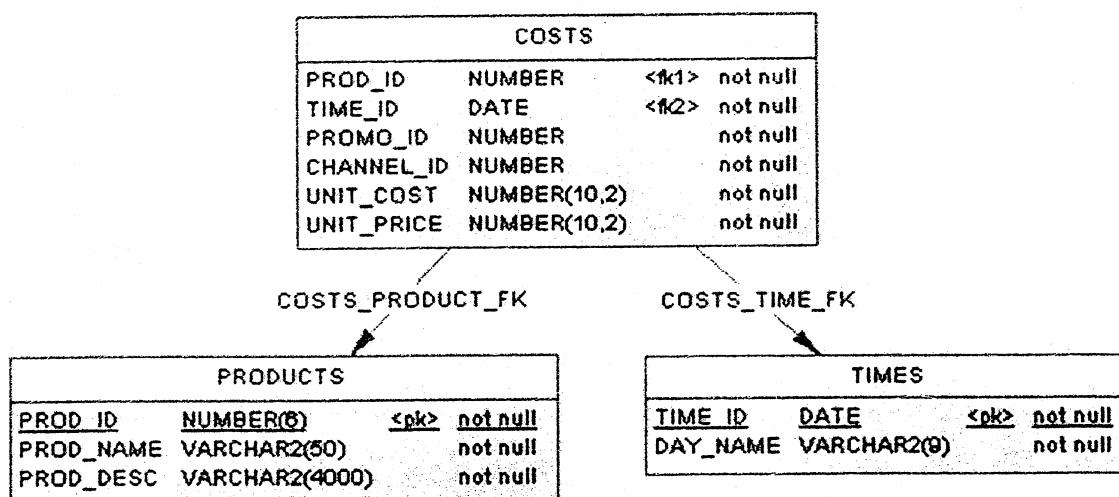


```
CREATE TABLE @EMP
  (EMP_ID NUMBER, EMP_NAME VARCHAR2 (12),
  EMP_ADD VARCHAR (30), EMP_DEP NUMBER, SALARY NUMBER);
```

The desired table is not created and an error message is returned. Which of the following is the cause of the error message?

- The data types used for every columns are inappropriate.
- The command used to create the table is inappropriate.
- The table name starts with a special character.
- The statement ends with a semi-colon (;).

Q No 14 Review the following ER diagram:



Which of the following queries will properly display the average unit\_cost (rounded to two decimal places) for a given product on a Saturday and Order the results by the PROD\_NAME column?

- SELECT a.prod\_name, ROUND(AVG(b.unit\_cost),2) FROM products a, costs b, times c WHERE a.prod\_id=b.prod\_id AND c.time\_id=b.time\_id AND c.day\_name='Saturday' GROUP BY a.prod\_name ORDER BY ROUND(AVG(b.unit\_cost),2);
- SELECT a.prod name, ROUND(AVG(b.unit cost),2) FROM products a, costs b, times c WHERE a.prod id=b.prod id AND c.time id=b.time id AND c.day name='Saturday' GROUP BY a.prod name ORDER BY a.prod name;
- SELECT a.prod\_name, AVG(b.unit\_cost) FROM products a, costs b, times c WHERE a.prod\_id=b.prod\_id AND c.time\_id=b.time\_id AND c.day\_name='Saturday' GROUP BY a.prod\_name ORDER BY a.prod\_name;
- SELECT a.prod\_name, ROUND(AVG(b.unit\_cost),2)FROM products a, costs b, times c WHERE a.prod\_id=b.prod\_idAND c.time\_id=b.time\_id AND c.day\_name='Saturday' ORDER BY a.prod\_name;



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E. `SELECT a.prod_name, ROUND(AVG(b.unit_cost),2) FROM products a, costs b, times c WHERE a.prod_id=b.prod_id AND c.time_id=b.time_id AND c.day_name='Saturday';`

Q No 15 The database contains a table named Employees\_Salary. The database also contains a public synonym named Emp\_Sal for the Employees\_Salary table. You want to remove the public synonym. Which of the following statements will he use to accomplish this?

- A. `DROP PUBLIC Emp_Sal;`
- B. `REMOVE SYNONYM Emp_Sal;`
- C. `DROP PUBLIC SYNONYM Emp_Sal;`
- D. `DROP SYNONYM Emp_Sal;`

Q No 16 MongoDB stores data as documents in a specific format representation called \_\_\_\_\_.

- A. JSON
- B. BSON
- C. XML
- D. no-sql

Q No 17 Which of the following types of queries can be used for highly scalable operational and analytic applications in MongoDB?

- A. Key-value queries
- B. Range queries
- C. Text Search queries
- D. All of the above

Q No 18

In MongoDB when you distribute data across multiple physical partitions it is called \_\_\_\_\_.

- A. sharding
- B. indexing
- C. caching
- D. distributed database

Q No 19 Which of the following data type is invalid in MongoDB?

- A. String
- B. Integer
- C. Array
- D. VARCHAR2

Q No 20 Which of the following MongoDB method replaces the existing document with the new document passed in method?

- A. `update()`
- B. `save()`
- C. `modify()`
- D. `insert()`



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Q No 21 Please review following code:

```
> db.library_collection.insert({
  _id: ObjectId(7df78ad8902c),
  title: 'MongoDB Overview',
  description: 'MongoDB is no sql database',
  by: 'SunBeam Institute',
  url: 'http://www.sunbeaminfo.com',
  tags: ['mongodb', 'database', 'NoSQL'],
  likes: 1000
})
```

Above code will insert document in library collection in order to insert multiple documents in single query which of the following query you need execute?

A. >db.library collection.insert([  
{  
  title: 'MongoDB Overview',  
  description: 'MongoDB is no sql database',  
  by: 'SunBeam Institute',  
  url: 'http://www.sunbeaminfo.com',  
  tags: ['mongodb', 'database', 'NoSQL'],  
  likes: 100  
},  
{  
  ...  
}]])

B. >db.library\_collection.insert(  
  {  
    title: 'MongoDB Overview',  
    description: 'MongoDB is no sql database',  
    by: 'SunBeam Institute',  
    url: 'http://www.sunbeaminfo.com',  
    tags: ['mongodb', 'database', 'NoSQL'],  
    likes: 100  
  },  
  {  
    ...  
  }  
)

C. >insert into library\_collection values(  
  {  
    title: 'MongoDB Overview',  
    description: 'MongoDB is no sql database',  
    by: 'SunBeam Institute',  
    url: 'http://www.sunbeaminfo.com',



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```
tags: ['mongodb', 'database', 'NoSQL'],
likes: 100
},
{
...
});
```

D. We cannot insert multiple documents in single query.

Q No. 22 Which of the following PL/SQL block executes automatically when an event occurs?

- A. Function
- B. Package
- C. Stored procedure
- D. Trigger

Q No 23 Obama works as an Application Developer for SunBeam Inc. The database contains two tables named Orders and Audit\_Records.

Obama is required to create a trigger on the Orders table to ensure that each time after a database user performs an UPDATE operation on the Orders table, his name and current date are inserted in the Audit\_Records table.

He also wants the user's name and current date to be inserted in the Audit\_Records table only once for an UPDATE operation, regardless of the number of rows updated by the UPDATE statement.

Which of the following statements will he use to accomplish this?

- A. CREATE OR REPLACE TRIGGER Order\_Trig AFTER UPDATE ON Orders FOR EACH ROW BEGIN INSERT INTO Audit\_Records (User\_Name, Date) VALUES (USER, SYSDATE); END;
- B. CREATE OR REPLACE TRIGGER Order Trig AFTER UPDATE ON Orders BEGIN INSERT INTO Audit Records (User Name, Date) VALUES (USER, SYSDATE);END;
- C. CREATE OR REPLACE TRIGGER Order\_Trig BEFORE UPDATE ON Orders BEGIN INSERT INTO Audit\_Records (User\_Name, Date) VALUES (USER, SYSDATE); END;
- D. CREATE OR REPLACE TRIGGER Order\_Trig BEFORE UPDATE ON Orders FOR EACH ROW BEGIN INSERT INTO Audit\_Records (User\_Name, Date) VALUES (USER, SYSDATE); END;

Q No 24 Which of the following statements is true about the cursor FOR loop?

- A. A cursor does not need to explicitly use the OPEN, FETCH, and CLOSE statements within a cursor FOR loop.
- B. A cursor needs to explicitly use the OPEN, FETCH, and CLOSE statements within a cursor FOR loop.



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- C. Parameters cannot be passed to a cursor in a cursor FOR loop.
- D. A cursor variable cannot be opened for a type-compatible query in a cursor FOR loop.

Q No 25 You work as an Application Developer for CDAC Inc. The company uses an Oracle database.

The database contains two tables named Employees and Audit\_Records. You are required to create a trigger on the Employees table in order to ensure that each time a database user performs an UPDATE operation on the Employees table, his name and current date are inserted in the Audit\_Records table. To accomplish this, you execute the following statement:

```
CREATE OR REPLACE TRIGGER Emp_Trig AFTER UPDATE ON Employees
BEGIN
    INSERT INTO Audit_Records (User_Name, Date) VALUES (USER, SYSDATE);
END;
```

A database user performs an UPDATE operation on the Employees table and updates five rows. How many times will his name and current date be inserted into the Audit\_Records table?

- A. 10
- B. 1
- C. 5
- D. The trigger will fail.

Q No 26 You work as an Application Developer for MKCL Inc. The company uses an Oracle database. The database contains a table named Employees. You have defined a database trigger named Raise\_Salary on the Employees table. You want to remove the trigger from the database. Which of the following SQL statements will you use to accomplish this?

- A. REMOVE TRIGGER Raise\_Salary;
- B. DELETE TRIGGER Raise\_Salary;
- C. DROP TRIGGER Raise Salary;
- D. ALTER TRIGGER Raise\_Salary REMOVE;

Q No 27 You work on a complex query for which you use multiple stored procedures. One of the stored procedures is required to pass a value to a higher-level stored procedure.

What will you use to pass a value from one stored procedure to another stored procedure?

- A. INOUT parameter
- B. Result statement
- C. IN parameter
- D. OUT parameter



# SAMPLE QUESTION PAPER

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Q No 28 Which of the following are valid trigger events for tables?

- i. AFTER STARTUP
- ii. AFTER DELETE
- iii. BEFORE INSERT
- iv. AFTER CREATE
- v. BEFORE LOGON

- A. Only i
- B. Only ii and iii
- C. Only iii and v
- D. Only i and ii

Q No 29 Which of the following statements about a package are true?

- A. A package has three parts.
- B. A package itself cannot be called or nested.
- C. A package itself can be called or nested.
- D. A package is loaded into the memory every time a package construct is called.

Q No 30 Under which of the following conditions is the use of an explicit cursor necessary?

- A. When any SQL data manipulation language (DML) statement is used
- B. When a query returns only one row
- C. When a query returns more than one row
- D. When a query does not return any row

Q No 31 The database contains a stored function named Get\_Sal. You want to re-create the Get\_Sal function with some modifications in the code. You also want to ensure that the privileges granted on the function remain intact. What will you do to accomplish the task with the least effort?

- A. Remove the Get\_Sal function by using the REMOVE FUNCTION statement and then create a new function with the same name by using the CREATE FUNCTION statement.
- B. Remove the Get\_Sal function by using the DELETE FUNCTION statement and then create a new function with the same name by using the CREATE FUNCTION statement.
- C. Remove the Get\_Sal function by using the DROP FUNCTION statement and then create a new function with the same name by using the CREATE FUNCTION statement.
- D. Create a new function with the same name by using the CREATE OR REPLACE FUNCTION statement.



## SAMPLE QUESTION PAPER

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Q No 32 We can assign a name to unnamed system exceptions using a Pragma called \_\_\_\_\_.

- A. EXCEPTION INIT
- B. ERROR\_INIT
- C. SQLERRM
- D. SQLCODE

Q No 33 The Employees table contains a column of the LONG datatype. Can you change the datatype of the column from LONG to BLOB?

- A. Yes, Using the ALTER TABLE statement.
- B. No, It is not possible.
- C. Can't say
- D. None of the above

Q No 33 What are the queries containing SET operators called?

- A. Subqueries
- B. Correlated subqueries
- C. GROUP BY queries
- D. Compound queries

Q No 34 What will be the outcome of the query given below?

SQL> SELECT 200+NULL+999 FROM dual;

- A. 200
- B. 999
- C. 1199
- D. NULL

Q No 35 What will be the result of the following query?

SQL> SELECT trunc(6902.92,-3) FROM dual;

- |                |            |
|----------------|------------|
| A. <u>6000</u> | C. 6700    |
| B. 6901        | D. 6901.00 |

Q No 36 Here are few statements about VARIANCE function in SQL.

- i. The function accepts multiple numeric inputs and returns variance of all the values
- ii. The function accepts a number column and returns variance of all column values including NULLs
- iii. The function accepts a number column and returns variance of all column values excluding NULLs

Chose the correct combination from the below options.

- |              |              |
|--------------|--------------|
| A. i and iii | C. <u>ii</u> |
| B. i and ii  | D. iii       |



# SAMPLE QUESTION PAPER

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- Q No 37 Index sort records into special data structure which known as \_\_\_\_\_.  
A. B-Tree  
B. AVL Tree  
C. B+ Tree  
D. All of the above
- Q No 38 What is the syntax of creating clustered index with index?  
A. CREATE CLUSTERED INDEX IndexName ON TableName (ColumnName);  
B. CREATE CLUSTERED INDEX ON TableName (ColumnName) WITH IndexName;  
C. CREATE CLUSTERED INDEX IndexName ON TableName.ColumnName  
D. Clustered Index can't be created
- Q No 39 A Table is 3NF if it is in 2NF and if it has no \_\_\_\_\_.  
A. functional dependencies  
B. transitive dependencies  
C. trivial functional dependency  
D. multivalued dependencies
- Q No 40 More than one materialized view cannot use a materialized view log  
A. True  
B. False





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## SAMPLE QUESTION PAPER

Course – DAC      Module - AWP



Q. No. 1

HTML Markup Language is set of Markup \_\_\_\_\_

Answer Choices

- A. Groups
- B. Sets

- C. Attributes
- D. Tags

Correct Answer : D

Q. No. 2

The HTML 5 element used to draw graphics via scripting is

Answer Choices

- A. <figcaption>
- B. figcaption>
- C. <mark>

- D. <canvas>
- E. <header>

Correct Answer: D

Q. No. 3

The <!DOCTYPE> declaration in HTML is

Answer Choices

- A. <!DOCTYPE html!>
- B. <!DOCTYPE html;>

- C. <!DOCTYPE html>
- D. <!DOCTYPE html:>

Correct Answer : C

Q. No. 4

HTML5 now supports Geolocation which can be used to find the location of the person browsing.

Answer Choices

- A. True

- B. False

Correct Answer: A

Q. No. 5

JavaScript can be used to \_\_\_\_\_

- A. Create HTML Elements
- B. Change HTML Elements

- C. Delete HTML Elements
- D. All of the above

Correct Answer: D

Q. No. 6

What is the output of following JavaScript code?

```
<script type="text/javascript">
var cst = new Array();
cst[0] = "Web Development";
cst[1] = "Application Development";
cst[2] = "Testing";
cst[3] = "ABC Computing";
document.write(cst[0,1,2,3]);
</script>
```

- A. Error
- B. ABC Computing
- C. Web Development

- D. Web Development , Application Development, Testing, ABC Computing

Correct Answer: B

Q. No. 7

JavaScript statements are executed by \_\_\_\_\_

- A. Compiler
- B. Server

- C. Browser
- D. JVM

Correct Answer: C



# SAMPLE QUESTION PAPER

Course – DAC      Module - AWP



Q. No. 8

```
<script type="text/javascript">
{
    document.write("<h1>This is heading </h1>");
    document.write("<p>This is a paragraph</p>");
    document.write("<p>This is another paragraph</p>");
}
```

Is above code error free?

A. Yes

B. No

Correct Answer: A

Q. No. 9

In JavaScript, which of the following is not a compound operator?

A. <=>

C. +=

B. >>==

D. ===

Correct Answer : A

Q. No. 10

What will be the output of following code?

```
var a="10";
var b=20;
var c=a + b;
document.write(c);
```

A. 30

C. 1020

B. 10

D. None of the

Correct Answer: C

Q. No. 11

Which of the following is not a valid way to begin and end a PHP code?

A. <% %>

D. <! !>

B. <? ?>

E. <?php ?>

C. <?= ?>

Correct Answer : D

Q. No. 12

What will the following script output?

```
<?php
$array = array (1, 2, 3, 5, 8, 13, 21, 34, 55);
$sum = 0;
for ($i = 0; $i < 5; $i++) {
    $sum += $array[$array[$i]];
}
echo $sum;
?>
```

A) 78

D) 5

B) 19

E) 0

C) NULL

Correct Answer: A



# SAMPLE QUESTION PAPER

Course – DAC      Module - AWP



*Prasanna Wadekar*

Q. No. 13

There are three types of arrays in PHP.

- A. Numeric array, String array, Multidimensional
- B. Numeric array, Associative array, Dimensional array
- C. Numeric array, Associative array, Multidimensional array
- D. Const array, Associative array, Multidimensional array

Correct Answer : C

Q. No. 14

In PHP fgets() is used to read a file one line at a time.

- A. True
- B. False

Correct Answer : A

Q. No. 15

In PHP in order to access MySQL database you will use:

- A. mysqlconnect() function
- B. mysql-connect() function
- C. mysql\_connect() function
- D. sql\_connect() function

Correct Answer: C

Q. No. 16

In PHP, the \_\_\_\_\_ function checks if the "end-of-file" (EOF) has been reached.

- A. f\_eof()
- B. f\_of()
- C. feofs()
- D. feof()

Correct Answer : D

Q. No. 17

In PHP, getdate() function returns ?

- A. An Integer
- B. A floating-point number
- C. An array
- D. A string
- E. A Boolean

Correct Answer : C

Q. No. 18

In PHP \_\_\_\_\_ which function is used to get number of records returned by SQL SELECT query.

- A. mysql\_num\_rows()
- B. odbc\_num\_rows()
- C. get\_mysql\_rows()
- D. None of the above

Correct Answer : A

Q. No. 19

In XML \_\_\_\_\_ is a term used about text data that will be parsed by the XML parser.

- A. CDATA
- B. PCDATA
- C. Parser
- D. None of the above

Correct Answer : B

Q. No. 20

How to specify the attributes with multiple values?

- A. <myelement myattribute="value1" myattribute="value2"/>



# SAMPLE QUESTION PAPER

Course – DAC      Module - AWP



- B. `<myelement myattribute="value1 value2"/>`
- C. `< myElement myAttribute="value1, value2"/>`
- D. attributes cannot have multiple values

Correct Answer: D

Q. No. 21

In XML, what is identified by a URI reference?

- A. XML namespace
- B. XSLT
- C. Schema
- D. DLT

Correct Answer : A

Q. No. 22

```
<!ELEMENT product (#PCDATA)>
<!ELEMENT product (quantity)>
<!ELMENET quantity (#PCDATA)>
```

What is wrong in above code?

- A. Two product elements are there with different definitions
- B. product element name is in lower case
- C. For numeric value like quantity PCDATA cannot be used.
- D. All of the above

Correct Answer : A

Q. No. 23

XML Schema consists of -

- A. Properties and methods
- B. elements and attributes
- C. structures and data
- D. tables and relationships

Correct Answer : B

Q. No. 24

```
<xsd:element name="price" type="xsd:integer" minOccurs='0' maxOccurs="unbounded"/>
```

The above declaration says price element can be used in XML file, minimum one time and maximum unlimited times. Which cardinality operator in a DTD says the same thing?

- A. ?
- B. +
- C. \*
- D. None of the above

Correct Answer : C

Q. No. 25

The XML Document Object Model (DOM) class is an in-memory representation of an XML document.?

- A. True
- B. False

Correct Answer : A



# SAMPLE QUESTION PAPER

Course – DAC      Module - AWP



Q. No. 26

```
xmlDoc=loadXMLDoc("products.xml");  
x=xmlDoc.getElementsByTagName("price");
```

This code will

- A. Return all price elements from products.xml file.
- B. Only first price element will be returned.
- C. Only last price element will be returned.
- D. All of the above.

Correct Answer : A

Q. No. 27

Which of the following is correct way to link XSLT to XML?

- i)      <?xml-stylesheet type="stylesheet" href="MyXSLT.xsl"?>
- ii)     <?xml-stylesheet type="text/xsl" href="MyXSLT.xsl"?>
- A. Only i
- B. Only ii
- C. Both i and ii
- D. None of i and ii

Correct Answer: B

Q. No. 28

XPath is used to -

- A. Find information in a XML document
- B. Design an XML document
- C. convert XML into HTML
- D. all of the above

Correct Answer: A

Q. No. 29

The document that is used by XSLT to indicate how to transform the elements of the XML document to another form is \_\_\_\_\_

- A. HTML Page
- B. DOCTYPE Procedure
- C. Stylesheet document
- D. Stored Procedures

Correct Answer : C

Q. No. 30

XPath is a W3C recommendation.

- A. True
- B. False

Correct Answer : A

Q. No. 31

The \_\_\_\_\_ approach of parsing XML document is event based.

- A. SAX
- B. DOM
- C. Both SAX and DOM
- D. JAXP

Correct Answer : A

Q. No. 32

SAX creates a tree-like-in-memory structure.

- A. True
- B. False

Correct Answer : B



# SAMPLE QUESTION PAPER

Course – DAC Module - AWP



Q. No. 33

Ajax based on

- A. JavaScript and XML
- B. VBScript and XML

C. JavaScript and Java

D. JavaScript and HTTP requests

Correct Answer : D

Q. No. 34

In Ajax readyState=3 means ?

A. XMLHttpRequest is interacting with the server.

C. XMLHttpRequest interaction is not yet started

B. XMLHttpRequest Interaction with the server is over.

D. None of the above

Correct Answer : A

Q. No. 35

The \_\_\_\_\_ protocol is termed as Common Messaging Protocol.

A. SOAP

C. UDDI

B. WSDL

D. CORBA

Correct Answer : A

Q. No. 36

WSDL stands for \_\_\_\_\_

A. Windows Services Description Language

C. Web Services Definition Language

B. Web Services Description Language

D. Web Security Description Language

Correct Answer : B

Q. No. 37

What script need to write for toggling fade?

A. fadeIn()

C. toggle()

B. fadeOut()

D. fadeToggle()

Correct Answer : D

Q. No. 38

jQuery is free and open source.

A. True

B. False

Correct Answer : A

Q. No. 39

\_\_\_\_\_ is a technique often used to attack data driven applications.

A. SQL Query

C. SQL Injection

B. PL SQL

D. SQL Views.

Correct Answer : C

Q. No. 40

XSS stands for \_\_\_\_\_

A. XML Site Scripting

B. Cross Site Scripting

C. Cross Source Scripting

D. Cross Structured Scripting

Correct Answer : B



# SAMPLE QUESTION PAPER

Course – DAC Module – Core Java (J2SE)



Q No 1 Which of the following statement is not correct?

- I Java is strongly type check language
- II Java is pure object oriented programming language
- III Java is type safe language
- iV Java is strictly type check language

- A. Only I
- B. II and III
- C. only III
- D. All of the above.

Q No 2 Given classes A, B, and C, where B extends A, and C extends B, and where all classes implement the instance method void doIt(). How can the doIt() method in A be called from an instance method in C?

- A. super.super.doIt();
- B. this.super.doIt(); (e)
- C. ((A) this).doIt();
- D. It is not possible.

Q No 3 Select the one correct answer?

- A. A JAR file can only contain one package.
- B. A JAR file can only be specified for use with the java command, in order to run a program.
- C. The classpath definition of the platform overrides any entries specified in the classpath
- D. The -d option is used with the java command, and the -D is used with the javac command.
- E. None of these

Q No 4 \_\_\_\_\_ is super class of all exception classes?

- A. Exception
- B. Throwable
- C. RuntimeException
- D. Error

Q No 5 parseInt() and parseLong() method throws \_\_\_\_\_ Exception ?

- A. ArithmeticException
- B. ClassNotFoundException
- C. ClassCastException
- D. NumberFormatException

Q No 6 Which one of the following interface, String class do not implement?

- A. Serializable
- B. Comparable
- C. Iterable
- D. CharSequence

Q No 7 Name of super class of StringIndexOutOfBoundsException class is \_\_\_\_\_

- A. String
- B. IndexOutOfBoundsException
- C. ArrayIndexOutOfBoundsException
- D. RuntimeException

Q No 8 What is the name of data member of class Vector which is used to store number of elements in the vector?

- A. length
- B. elements
- C. elementCount
- D. capacity



# SAMPLE QUESTION PAPER

Course – DAC

Module – Core Java (J2SE)



Q No 9 What will be the output of the program?

```
TreeSet map = new TreeSet();
map.add("one");
map.add("two");
map.add("three");
map.add("four");
map.add("one");
```

```
Iterator it = map.iterator();
while (it.hasNext() )
{
    System.out.print( it.next() + " " );
}
```

- A. one two three four
- B. four three two one
- C. four one three two
- D. one two three four one

Q No

A.

B.

Q No

cl

Q No 10 What will be the output of the program?

```
import java.util.*;
class Program {
    public static void main (String[] args) {
        Object x = new Vector().elements();
        System.out.print((x instanceof Enumeration)+",");
        System.out.print((x instanceof Iterator)+",");
        System.out.print(x instanceof ListIterator);
    }
}
```

- A. false false false
- B. false false true
- C. false true false
- D. true false false

Q No

A. By

it

B. By

C. A t

D. Two

Q No

A.

B.

C.

D.

Q No

A.

B.

C.

D.

Q No 11 Which of the following are true statements?

- I) The Iterator interface declares only three methods: hasNext, next and remove.
  - II) The ListIterator interface extends both the List and Iterator interfaces.
  - III) The ListIterator interface provides forward and backward iteration capabilities.
  - IV) The ListIterator interface provides the ability to modify the List during iteration.
  - V) The ListIterator interface provides the ability to determine its position in the List.
- A. II, III, IV and V
  - B. I, III, IV and V
  - C. III, IV and V
  - D. I, II and III

Q No

witho

A.

B.

C.

D.

Q No 12 \_\_\_\_\_ directory contains files for compiling native methods?

- A. src
- B. lib
- C. jre
- D. include

Q No

class.

A.

B.



# SAMPLE QUESTION PAPER

Course – DAC      Module – Core Java (J2SE)

Q No 13 Which of these is the interface of legacy?

- A. Map
- B. Enumeration
- C. HashMap
- D. Hashtable

Q No 14 What is the output of this program? class Program

```
class Program {
    public static void main(String args[]) {
        Thread t = Thread.currentThread();
        System.out.println(t);
    }
}
```

- A. Thread[5 main]
- B. Thread[main 5]
- C. Thread[main 0]
- D. Thread[main 5 main]

Q No 15 Which of these statements is incorrect?

- A. By multithreading CPU's idle time is minimized, and we can take maximum use of it
- B. By multitasking CPU's idle time is minimized, and we can take maximum use of it
- C. A thread can exist only in two states, running and blocked
- D. Two thread in Java can have same priority

Q No 16 To kill a thread at any stage \_\_\_\_\_ method is used.

- A. kill
- B. stop
- C. destroy
- D. dead

Q No 17 Default priority of thread in java is

- A. Depends on priority of parent thread.
- B. NORM\_PRIORITY
- C. NORMAL
- D. NORMAL\_PRIORITY

Q No 18 What will be the result of invoking the wait() method on an object without ensuring that the current thread holds the lock of the object?

- A. The code will fail to compile.
- B. Nothing special will happen.
- C. An exception will be thrown if the wait() method is called while the current thread does not hold the lock of the object.
- D. The thread will be blocked until it gains the lock of the object.

Q No 19 To create directory we should use \_\_\_\_\_ method of java.io.File class.

- A. mkdir
- B. mkdir
- C. createDir
- D. newDir



Prasanna wadekar

## SAMPLE QUESTION PAPER

Course – DAC Module – Core Java (J2SE)



Q No 20 While Comparing BufferedWriter and FileWriter, Which capability exist as a method in only one of two?

- A. Closing the stream
- B. Flushing the stream
- C. Writing to the stream
- D. Writing a line separator to the stream
- E. None of these

Q No 21 We can persist state of an object using \_\_\_\_\_ method.

- A. serialie
- B. save
- C. writeObject
- D. write

Q No 22 The \_\_\_\_\_ parameter is used to specify the applet class name

- A. classname
- B. code
- C. name
- D. codebase

Q No 23 HTML based java documentary help can be accessed using \_\_\_\_\_

- A. javadoc
- B. javahelp
- C. javamanual
- D. None of these

Q No 24 In \_\_\_\_\_ component we can select multiple items.

- A. Option Button
- B. Choice
- C. List
- D. None of these

Q No 25 Default layout manager for Panel is

- A. FlowLayout
- B. BorderLayout
- C. GridLayout
- D. None

Q No 26 \_\_\_\_\_ attribute specifies the directory that contains the applet's code.

- A. src
- B. code
- C. codebase
- D. dir

Q No 27 paint is method of which of the following class

- A. Component
- B. Container
- C. Window
- D. Frame

Q No 28 setBackground is a method of which of the following class?

- A. Applet
- B. Container
- C. Graphics
- D. Component

Q No 29 \_\_\_\_\_ method of AppletContext interface can be used to transferring control to the other html page

- A. appletviewer
- B. init
- C. showDocument
- D. showApplet

Q No 30 invoke is method of which of the following

- A. Method
- B. Class
- C. Proxy
- D. None of these

# SAMPLE QUESTION PAPER

Course – DAC      Module – Core Java (J2SE)

- Q No 31 Following statement contains \_\_\_\_\_ type of wild card.  
`Node<? super Integer > node = new Node<? super Integer >();`  
 A. Unbounded wild card      C. Upper bounded wild card  
 B. Lower bounded wild card      D. None of these
- Q No 32 Which of these class is used to encapsulate IP address and DNS?  
 A. DatagramPacket      C. InetAddress  
 B. URL      D. ContentHandler
- Q No 33 Which of these is a bundle of information passed between machines  
 A. MIME      C. Datagrams  
 B. Cache      D. DatagramSocket
- Q No 34 Which of the following extends Number class?  
 A. Boolean      D. All of the above  
 B. Character      E. None of the above  
 C. Void
- Q No 35 When one end of TCP connection closes it by making a system call, it goes into \_\_\_\_\_ state.  
 A. TIME\_NOTIFY      C. TIME\_SLEEP  
 B. TIME\_WAIT      D. TIME\_BLOCK
- Q No 36 How can you explicitly ensure that the memory allocated by an object is freed?  
 A. By invoking the free method on the object.  
 B. By calling System.gc() method.  
 C. By setting all references to the object to new values (say null).  
 D. Garbage collection cannot be forced. The programmer cannot force the JVM to free the memory used by an object.
- Q No 37 Which of the following are legal array declarations?  
 I `int i[5][];`  
 II `int i[][];`  
 III `int []i[];`  
 IV `int i[5][5];`  
 V `int[][] a;`  
 A. I and IV  
 B. II and V  
 C. II, III and V  
 D. I, II, III and V
- Q No 38 What is the difference between application and applet?  
 A. An application can in general be trusted whereas an applet can't.  
 B. An applet may be executed in a browser environment.  
 C. An applet is not able to access the files of the computer it runs on  
 D. A and B  
 E. A B and C



## SAMPLE QUESTION PAPER

Course – DAC

Module – Core Java (J2SE)



Q No 39 Select the correct statement

- I We can declare static methods inside Inner class.
  - II We can access non static members directly inside static nested class
  - III We cannot access non static members directly inside Inner class
  - IV We can declare final field as static inside Inner class
- A. only IV
  - B. II and III
  - C. I, II and III
  - D. None of these

Q No 40 Select the one correct answer :

- A. Private methods cannot be overridden in subclasses.
- B. An overriding method can declare that it throws checked exceptions that are not thrown by the method it is overriding.
- C. The parameter list of an overriding method can be a subset of the parameter list of the method that it is overriding.
- D. The overriding method must have the same return type as the overridden method.



# SAMPLE QUESTION PAPER

Course – DAC      Module – Ent. Java (J2EE)



Q No 1 Around Advice can access the return value of business method and it can modify the value and it can return a different value back to the client, as return type is Object,

- A. After advice it is not possible to make changes
- B. if return type is void
- C. A & B
- D. after advice also we can modify the value

Q No 2 Spring AOP supports various types of advices

- i Before advice
- ii After returning advice
- iii After throwing advice
- iv Around advice
- v After

- A. i,ii,iii
- B. iv,v
- C. i & v
- D. A & B

Q No 3 In database transaction processing ACID property refers to the

- A. Atomicity, Consistency, Isolation, Declaration respectively.
- B. Atomicity, Consistency, Isolation, Durability respectively.
- C. Atomocity, Consistency, Insertion, Deletetion respectively.
- D. Automatic, Concurrent, Intermediate, Durable respectively.

Q. No 4 Static initializer blocks are used to initialize static fields which can be overcome in ejb using methods

- A. ejbCreate()
- B. setSessionContext()
- C. setEntityContext()
- D. all of the above

Q No 5 The ----- is a method used to locate and load an Entity Bean into the container.

- A. ejbFind()
- B. ejbSearch()
- C. ejbFindByPrimaryKey()
- D. none of the above

Q. No 6 A ----- provides the ability for any Enterprise Information System (EIS) to plug into any EJB server which supports the Connector architecture.

- A. connector
- B. container
- C. server
- D. ejb



## SAMPLE QUESTION PAPER

Course – DAC      Module – Ent. Java (J2EE)



- Q. No 7 which of the following is false statement mdb
- A. are instantiated by the container as required
  - B. implementation of `onMessage()` contains the business logic required to process the message.
  - C. message beans have home or remote interfaces
  - D. message driven beans implement the `javax.jms.MessageListener`
- Q. No 8 ----- refers to the ability of multiple load-balanced web servers to share session and entity data. It is a major feature of web application servers.
- A. segmentation
  - B. Clustering
  - C. activation
  - D. passivation
- Q No 9 Which of the following is not true about EJB
- A. The EJB specification specifically states that the enterprise bean is not allowed to use thread primitives.
  - B. The container is responsible for managing concurrent access to beans at runtime.
  - C. We can change the transaction isolation level in the middle of transaction.
  - D. A & B
- Q No 10 A Component Transaction Monitor (CTM) is an application server that uses a server-side component model, which facilitates
- A. to provide service managing transactions, security, and concurrency.
  - B. distributed obj architectures
  - C. provide facilities for object persistence.
  - D. All of the above
- Q No 11 What is true about business methods
- A. mandating non-final business methods allows container developers to implement their EJB container via inheritance
  - B. business methods should not be declared as final
  - C. Nobody can generate class that extends bean if method is final
  - D. A & B
- Q.No 12 A JAAS-based application implements the -----interface so that it can interact with users to enter specific authentication data, such as user names or passwords, or to display error and warning messages
- A. `javax.security.CallbackHandler`
  - B. `javax.security.auth.CallbackHandler`
  - C. `javax.security.auth.callback.CallbackHandler`
  - D. `javax.security.auth.callback.PostbackHandler`



# SAMPLE QUESTION PAPER

Course – DAC Module – Ent. Java (J2EE)



Q No 13 The ----- element configures a set of beans that are instances of Map. whereas the ----- element configures a set of beans that are instances of List.

- A. map-element , list element
- B. map-entries , list-entries
- C. map, list
- D. B or C

Q.No 14 The JSF life cycle handles

- A. initial requests and postbacks requests
- B. initial requests
- C. postbacks requests
- D. none of the above

Q.No 15 JavaServer Faces technology provides a rich architecture for managing

- A. component state & processing component data
- B. validating user input & handling events
- C. A & B
- D. None of the above

Q No 16 ----- based life cycle invocation makes servlet based web-application scalable

- A. session
- B. Web based
- C. Thread
- D. n-tire

Q No 17 Struts2 is a

- A. push-MVC framework
- B. pull-MVC framework
- C. A & B
- D. None of the above

Q No 18 Which of the following is true?

- A. If the ActionContextCleanUp filter is present, the FilterDispatcher will not clean up the ThreadLocal ActionContext.
- B. If the ActionContextCleanUp filter is not present, the FilterDispatcher will cleanup all ThreadLocals.
- C. If the ActionContextCleanUp filter is not present, the FilterDispatcher will not cleanup all ThreadLocals.
- D. A & B

Q No 19 What is not true about ValueStack

- A. the storage area where the application data is stored by Struts2 for processing the client requests.
- B. the data is stored in ActionContext objects
- C. interceptor used to manipulate data stored
- D. the action context objects use ThreadLocal to have values specific to the particular request thread.



# **SAMPLE QUESTION PAPER**

Course – DAC      Module – Ent. Java (J2EE)



- Q No 20 struts.devMode gives which one of the following benefits
- A. resource bundle reload on every request
  - B. struts.xml or any configuration files can be modified without restarting or redeploying the application
  - C. The error occurs in the application will be reported, as oppose to production mode.
  - D. all of the above
- Q No 21. An abstract package usually defines inheritable components except
- A. interceptor
  - B. different interceptor stacks
  - C. actions
  - D. result types
- Q No 22 The filter chain includes
- A. Action filter
  - B. FilterDispatcher
  - C. ActionInvocation
  - D. Action Mapping
- Q No 23 Field and Non Field / Plain are the scope of
- A. Action
  - B. Field
  - C. Validator
  - D. package
- Q No 24 Criteria in hibernate is a
- A. simple yet powerful API
  - B. which is used to retrieve entities
  - C. Entries are retrieved through criteria object composition.
  - D. all of the above
- Q No 25 What are two types of collections are available in hibernate ?
- A. Sorted Collection & Order Collection
  - B. Sorted Collection & List Collection
  - C. Sorted Collection & Query Collection
  - D. none of the above
- Q No 26 Which of the following are ORM levels in hibernate ?
- A. Pure Relational
  - B. Light Object Mapping
  - C. Medium Object Mapping
  - D. Full Object Mapping
  - E. all of the above





# SAMPLE QUESTION PAPER

Course – DAC      Module – Ent. Java (J2EE)



Q No 27 javax.servlet.ServletRequestListener and ServletRequestEvent monitored and implemented for event

- A. A servlet request has started being processed by web components
- B. Attribute added, removed, or replaced
- C. Creation, invalidation, activation, passivation, and timeout
- D. Initialization and destruction

Q No 28 Which one is the default transaction factory in hibernate?

- A. CMTTransactionFactory
- B. JTATransactionFactory
- C. JDBCTransactionFactory
- D. DEFAULTTransactionFactory

Q No 29 Version property is used in hibernate

- A. to know whether an object is in transient state or in detached state.
- B. to know whether an object is in transient state and detached state.
- C. to know whether an object is in transient state
- D. Detached state.

Q No 30 What are the inheritance models are of hibernate

- A. Tables Per Concrete Class
- B. Table per class hierarchy
- C. Table per sub-class
- D. all of the above

Q No 31 Light Object Mapping is one of the levels of ORM quality

- A. in which all entities are represented as classes and they are mapped manually.
- B. in which all entities are represented as hbm.xml and they are mapped manually.
- C. A & B
- D. none of the above

Q No 32 Hibernate associations are

- A. unidirectional
- B. bidirectional
- C. relate to container management persistence
- D. none of the above

Q No 33 Which one of the following fetching strategies are available in hibernate

- A. Join Fetching
- B. Select Fetching
- C. A & B
- D. C & Batch Fetching

Q No 33 ----- provides tools for development of efficient and robust distributed, web based solutions

- A. JMS
- B. JDK
- C. JMX
- D. JAS



# SAMPLE QUESTION PAPER

Course – DAC      Module – Ent. Java (J2EE)



- Q No 34 Contents and encoding of response can be set using
- A. `<%@ page contentType="text/html; charset=UTF-8" %>`
  - B. `<%@ page contenttype="text/html; charset=UTF-8" %>`
  - C. `<%@ page content="text/html; charset=UTF-8" %>`
  - D. `<%@ page content type="text/html" %>`
- Q No 35 In addition to scope object attributes, shared resources include
- A. in-memory data (such as instance or class variables)
  - B. external objects such as files
  - C. Database connections and network connections.
  - D. all of the above
- Q No 36 The object that implements a custom tag is called
- A. tag handler
  - B. Scripting Element
  - C. TagListener
  - D. A tag File
- Q No 37 An instance of the servlet does not exist, the web container in sequence of
- i) Loads the servlet class.
  - ii) Initializes the servlet instance by calling the init method.
  - iii) Creates an instance of the servlet class.
- A. i, ii, iii
  - B. i, iii, ii
  - C. ii, iii, i
  - D. ii, i, iii
- Q No 38 Which of the following are action tags in JSP ?
- A. `jsp:include` and `jsp:forward`
  - B. `jsp:useBean`
  - C. `jsp:setProperty` and `jsp:getProperty`
  - D. `jsp:plugin`
  - E. All of the above
- Q No 39 ----- is a small bit of text information which a web server sends to a browser and which browsers returns when it visits the same site again
- A. url
  - B. request parameter
  - C. cookie
  - D. sessionId
- Q No 40 How to implement Thread Safe in JSP ?
- A. `<%@ page ThreadSafe="true" %>`
  - B. `<%@ page isThreadSafe="false" %>`
  - C. `<%@ page ThreadSafe="current" %>`
  - D. `<%@ import ThreadSafe="true" %>`



*Prasanna Wadekar*

## SAMPLE QUESTION PAPER

Course – DAC      Module – MS.NET



Q No 1 We can check existence of .NET framework by looking towards \_\_\_\_\_ file.

- A. clr.dll
- B. mscorlib.dll
- C. mscorwks.dll
- D. mscorlib.dll

Q No 2 Which method is used to create proxy object using metadata in Remoting ?

- A. Activator.CreateInstance
- B. Activator.GetInstance
- C. Activator.GetObject
- D. Activator.NewInstance

Q No 3 Which of the following is base class for delegate type?

- A. System.Object
- B. System.Delegate
- C. System.UnicastDelegate
- D. System.MulticastDelegate

Q No 4 \_\_\_\_\_ provides object, which describes assemblies and its contents

- A. Reflection
- B. Attribute
- C. Metadata
- D. Manifest

Q No 5 To provide control access to the custom attribute we should use which of the following attribute

- A. Attribute
- B. ObsoleteAttribute
- C. ConditionalAttribute
- D. AttributeUsage

Q No 6 Identify the interface listed below which is not declared in System.Collections namespace?

- A. IDictionary
- B. IDictionaryEnumerator
- C. IDictionaryComparer
- D. IEnumerable

Q No 7 Form class is derived from \_\_\_\_\_?

- A. Container
- B. Control
- C. ContainerControl
- D. Component



# SAMPLE QUESTION PAPER

Course – DAC      Module – MS.NET



Q No 8 \_\_\_\_\_ folder Contains a collection of files that define the appearance of ASP.NET Web pages and controls.

- A. Theme
- B. Themes
- C. App Themes
- D. App\_LocalResources

Q No 9 A postback occurs when:

- A. End User submits information in Form to the server
- B. user access the web page using URL
- C. Server posts a form to the client.
- D. Both A and B
- E. All of the above.

Q No 10 Which is the attribute of Validation Control used for applying Validation to particular TextBox control of the Form ?

- A. Validate
- B. ValidateControl
- C. ControlToBind
- D. ControlToValidate

Q No 11 You develop a Web application that contains two master pages. You need to dynamically set the master page when a user views pages in the application. What should you do?

- A. Set Page.MasterPageFile in the Page's Page\_Init event.
- B. Set Page.MasterPageFile in the Page's OnInit event.
- C. Set Page.MasterPageFile in the Page's Page\_Load event.
- D. Set Page.MasterPageFile in the Page's Page PreInit event.

Q No 12 To create multifile assembly, which of the following tool we should use?

- A. ngen.exe
- B. csc.exe
- C. ilasm.exe
- D. al.exe

Q No 13 Which of the following is TRUE about Windows Authentication in ASP.NET?

- A. Automatically determines role membership
- B. Role membership determined only by user programming
- C. ASP.NET does not support Windows Authentication
- D. All of the Above

Q No 14 Which control is required for every page that have AJAX Extensions for ASP.NET

- A. UpdatePanel
- B. ScriptManager
- C. ContentPanel
- D. None



## SAMPLE QUESTION PAPER

Course – DAC      Module – MS.NET



- Q No 15 Which interface defines the basic extension methods for LINQ?
- A. IComparable< T>
  - B. IList
  - C. IEnumerable
  - D. IQueryable< T>
- Q No 16 Which of the following method of the command object is best suited when you have aggregate functions in a SELECT statement?
- A. ExecuteScalar
  - B. ExecuteReader
  - C. ExecuteNonQuery
  - D. NONE
- Q No 17 Which of the following is the method provided by the DataSet object to generate XML file?
- A. ReadXML
  - B. WriteXML
  - C. GetXML
  - D. All of above
- Q No 18 WSDL stands for \_\_\_\_\_.
- A. Web Services Description Language
  - B. Web Services Designing Language
  - C. Windows Services Description Language
  - D. None of these
- Q No 19 \_\_\_\_\_ binding provides secure binding for peer-to-peer environment and network applications.
- A. NetPeerTcpBinding
  - B. NetTcpBinding
  - C. BasicHttpBinding
  - D. WSDualHttpBinding
- Q No 20 \_\_\_\_\_ is a formal agreement between a service and a client that abstractly describes the data to be exchanged.
- A. service contract
  - B. data contract
  - C. message contract
  - D. operation contract
- Q No 21 What is the default authentication mode for IIS?
- A. Windows
  - B. Anonymous
  - C. Basic Authentication
  - D. None



# SAMPLE QUESTION PAPER

Course – DAC Module – MS.NET



- Q No 22 Which is the incorrect statement against Controller conventions followed in ASP.NET MVC?
- A. Controller must be suffixed with "Controller"
  - B. Each Controller class implement IController interface
  - C. Each Controller Action method must return Action Result or void
  - D. Action method of Controller must have out/ref parameters
- Q No 23 What is the default scope for instance Variable declared in class ?
- A. protected
  - B. internal
  - C. protected internal
  - D. private
- Q No 24 Linux version of Silverlight is called
- A. MonoLight
  - B. Silverlight
  - C. Moonlight
  - D. None of these
- Q No 25 Which is the default method that will be called on a controller if one is not explicitly specified in Asp.Net MVC ?
- A. Home
  - B. Index
  - C. WelCome
  - D. None of these
- Q No 26 \_\_\_\_\_ referes to formatting data based on locale?
- A. Globalization
  - B. Localization
  - C. Both A and B
  - D. None of these
- Q No 27 Which DLL translates XML to SQL in IIS?
- A. sqlisapi.dll
  - B. sqlxml.dll
  - C. lisxml.dll
  - D. sqliis.dll
- Q No 28 Which is the tool used for deploying Assembles in Global Assembly Cache?
- A. Reg\$vr.exe
  - B. GacUtil.exe
  - C. Gac\$vr32.exe
  - D. GacMgr.exe
- Q No 29 what is default time of expire session?
- A. 20 minute
  - B. 40 minute
  - C. 100 minute
  - D. 1200 minute
- Q No 30 You have created a new ASP.Net web application. You build the solution in Visual Studio .Net. You notice that there is a /bin directory under the virtual directory. Which type of files should you place in this location?
- A. Assemblies that are used on only this one web application
  - B. Assemblies that are used on web sites on only this web server machine.
  - C. Assemblies that are used by web applications anywhere on the network
  - D. None of the above



# SAMPLE QUESTION PAPER

Course – DAC      Module – MS.NET



Q No 31 The ASP.NET directive that lets you cache different versions of a page based on varying input parameters, HTTP headers and browser type is \_\_\_\_\_.

- A. @CacheOutput
- B. @PageCache
- C. @CacheAll
- D. @OutputCache

Q No 32 Which one of the following namespaces contains the definition for IDbConnection?

- A. System.Data.Interfaces
- B. System.Data.Common
- C. System.Data
- D. System.Data.Connection

Q No 33 In the Entity Framework, which is the primary object that you use to query and modify data?

- A.ObjectContext
- B. EntityObject
- C. DataContext
- D. None Of these

Q No 34 Which of the following class we should extend to access it outside the appdomain?

- A. MarshalByRefObject
- B. MarshalByReferenceObject
- C. MarshalByValObject
- D. MarshalByValueObject

Q No 35 \_\_\_\_\_ class represents a control consisting of a movable bar that divides a container's display area into two resizable panels.

- A. Splitter
- B. Container
- C. ContextMenuStrip
- D. SplitContainer

Q No 36 Which of the following layout is not available in WPF?

- A. StackPanel
- B. Canvas
- C. Grid
- D. Table

Q No 37 WrapPanel Class comes under \_\_\_\_\_ assembly.

- A. PresentationCore
- B. PresentationFramework
- C. milcore
- D. None of above

Q No 38 \_\_\_\_\_ is super class of enum

- A. System.ValueType
- B. System.Object
- C. System.Enum
- D. enum do not extend any class.

Q No 39 With the help of \_\_\_\_\_, we can pass initialized variable to the function by reference.

- A. ref
- B. out
- C. params
- D. var

Q No 40 Array.Sort method implicitly use \_\_\_\_\_ algorithm to sort element.

- A. Merge sort
- B. Heap Sort
- C. Insertion Sort
- D. Quick sort







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Q No 01 Deliverable for software Project is

- A. Source Code
- B. Design Documents
- C. Requirement Documents and Test Plans
- D. All of the above

Q No 02 Scoping is done during,

- A. Proposal Stage
- B. Requirements gathering stage
- C. Design Stage
- D. Coding Stage

Q No 03 Which SDLC Model is best suited for doing 'Proof of Concept' when technologies are new and we are not sure whether they work well with each other.

- A. Waterfall Model
- B. Incremental Model
- C. Prototype Model
- D. Spiral Model

Q No 04 While gathering requirements, Progressive elaboration is to be done in which sequence?

- A. Needs, Software Requirements, Features
- B. Features, Needs, Software Requirements
- C. Needs, Features, Software Requirements
- D. None of the above

Q No 05 WBS is,

- A. Describes Stakeholder needs, High-level product description, Project Justification
- B. Describes Project Objectives, Product Requirements, Deliverables, Acceptance criteria
- C. Starts with Phase level Deliverables; decomposes to the work package level
- D. All of the above

Q No 06 While gathering the requirements on OO way (using RUP UML), the very first thing we should do it

- A. Start gathering functional requirements
- B. List down all the Users of the System (called as Actors)
- C. Start gathering non-functional requirements
- D. Create Test plan

Q No 07 What is the solution to "Yes-But Syndrome" in requirements gathering

- A. Improve technical skills
- B. Seek customer feedback early
- C. Learn a tool for requirements
- D. None of the above

Q No 08 In case of Bank, what will be the relationship between "Closing of Account" use case and "Withdrawal" Use case.

- A. Uses
- B. Includes
- C. Extends
- D. None of the above

Q No 09 In case of Bank, what will be the relationship between "Opening a Saving Account" use case and "Opening Account" Use case.

- A. Uses
- B. Includes
- C. Extends
- D. None of the above



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- Q No 10 Which of the following statement is true for Object Oriented languages:
- A. Use Top-Down approach for Design
  - B. More development time is required
  - C. Model real-world well (natural way of programming)
  - D. Less re-use
- Q No 11 Type/Types of Software Maintenance is/are,
- A. Corrective
  - B. Adaptive
  - C. Perfective
  - D. All of the above
- Q No 12 Memory leakage testing is of which type?
- A. White Box Testing
  - B. Black Box Testing
  - C. Boundary Value
  - D. None of the above
- Q No 13 ISO Certification series has the following certification(s),
- A. ISO 9001
  - B. ISO 9002
  - C. ISO 9003
  - D. All of the above
- Q No 14 CMM is applicable to which industry?
- A. Information Technology
  - B. Civil
  - C. Mechanical
  - D. All of the Above
- Q No 15 "Managed" is which level of CMM?
- A. 2
  - B. 3
  - C. 4
  - D. 5
- Q No 16 Checking Items and Rejecting Defected Items is,
- A. Inspection
  - B. Quality Control
  - C. Quality Assurance
  - D. TQM
- Q No 17 In what stages of software could there be risks?
- A. Requirement Management
  - B. Design
  - C. Coding
  - D. All of the above
- Q No 18 "Is this what customer wanted" is,
- A. Verification
  - B. Validation
  - C. Boundary Value Analysis
  - D. All of the above
- Q No 19 Statement of Work is,
- A. Describes Stakeholder needs, High-level product description, Project Justification
  - B. Describes Project Objectives, Product Requirements, Deliverables, Acceptance criteria
  - C. Starts with Phase level Deliverables; decomposes to the work package level
  - D. All of the above



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- Q No 20 For gathering requirements, you need to
- A. Develop Domain expertise
  - B. Need to have the skill to ask right questions
  - C. Need to do understand customer's business problem/opportunity
  - D. All of the above
- Q No 21 What is the solution for "User-Developer Syndrome",
- A. Gain Tool Knowledge
  - B. Gain more technical knowledge
  - C. Gain Domain knowledge
  - D. None of the above
- Q No 22 Following development based on Requirements,
- A. Design
  - B. Test plans
  - C. Estimations
  - D. All of the above
- Q No 23 Which of the following is true for Activity diagrams?
- 1 They are used to capture the flow in a use case
  - 2. It is a set of scenario that describe an interaction between a user and a system
  - 3. They are similar to flow charts
  - 4. It covers activities of classes
- A. 1, 2
  - B. 1, 2, 3
  - C. 1, 3
  - D. All of the above
- Q No 24 In case of ATM, what will be the relationship between "Authentication" use case and "Check Balance" Use case.
- A. Uses
  - B. Includes
  - C. Extends
  - D. None of the above
- Q No 25 Types of Actors is/are,
- A. Roles
  - B. Hardware
  - C. Software
  - D. All of the Above
- Q No 26 Pareto's rule says,
- A. 80% value comes from 20% features
  - B. 70% value comes from 30% features
  - C. 60% value comes from 40% features
  - D. 50% value comes from 50% features
- Q No 27 Build Scripts are useful for,
- A. Requirements Gathering
  - B. Design
  - C. Software compiling and packaging
  - D. Testing
- Q No 28 Main Objective of Software Testing is to,
- A. Prove that the software does what it is supposed to do
  - B. To check non-functional defects
  - C. To find maximum number of defects.
  - D. None of the above



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Q No 29 For Critical documents like estimation and proposal, which of the following review is best,

- A. Peer review
- B. senior review
- C. Group review
- D. Brainstorming

Q No 30 White box testing is also called,

- A. Glass box testing
- B. Structural testing
- C. Clear box testing
- D. All of the above

Q No 31 Acceptance requirements that a system should meet is

- A. Usability
- B. Understandability
- C. Functionality
- D. Enhancement

Q No 32 A critical path network diagram does not.

- A. Calculate earned-value
- B. Identify the particularly important activities
- C. Calculate the duration of the whole project
- D. Help determine the amount of float

Q No 33 Which of the following is the lowest risk, least extent of change?

- A. Business process improvement.
- B. Business process re-engineering.
- C. Incremental change.
- D. Business process automation.

Q No 34 Project risk factor is considered in ?

- A. Spiral Model
- B. Waterfall Model
- C. Prototyping Model
- D. Iterative enhancement Model

Q No 35 The cost of software engineering (if we use waterfall model) includes approximately ..... of development costs and ..... of testing costs.

- A. 50%, 50%
- B. 40%, 60%
- C. 60%, 40%
- D. 80%, 20%

Q No 36 The choice of the Software Development Life Cycle Model to be followed for a project depends on

1. Initial Clarity of Requirements
  2. Size of the Project
  3. Time Frame of the Project
  4. Clarity on Technical Issues
- A. 1, 2 & 3 only
  - B. 1, 2 & 4 only
  - C. 1, 2, 3 & 4
  - D. 1 & 4 only



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Q No 37 The majority of the lifetime of a program is spent in the \_\_\_\_\_ phase

- A. Maintenance
- B. Analysis
- C. Design
- D. Testing

Q No 38 Balancing of DFD means

- A. Conservation of inputs & outputs at various levels
- B. Sub dividing a process into smaller sub processes
- C. Labeling of all data items
- D. Allowing data flows to take place only to or from processes

Q No 39 The first reverse engineering activity involves seeking to understand

- A. data
- B. processing
- C. user interfaces
- D. none of above

Q No 40 \_\_\_\_\_ ensures that a set procedure is followed to make any changes to the software

- A. Risk Management
- B. COCOMO
- C. Software Configuration Management
- D. Transform Analysis

